

Tutorial: exam-style practice

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Part A: 10 multiple-choice questions (1 mark each). Part B: 5 structured questions (27 marks). Same format as the final exam.

Answer Part A by circling one option. Answer Part B in full, showing your working. A periodic table and the answer key are at the back.

Part A · Multiple choice

A1 Topic: Chemical Reactions II CR2-LO1

Which statement about an endothermic reaction is correct?

- A. ΔH is negative and the surroundings get warmer
- B. ΔH is positive and heat is absorbed from the surroundings
- C. Bond forming releases more energy than bond breaking absorbs
- D. It can never happen at room temperature

A2 Topic: Chemical Reactions II CR2-LO1

$\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$, $\Delta H = +180.6 \text{ kJ}$. What is ΔH for $4\text{NO} \rightarrow 2\text{N}_2 + 2\text{O}_2$?

- A. $+361.2 \text{ kJ}$
- B. -180.6 kJ
- C. -361.2 kJ
- D. $+90.3 \text{ kJ}$

A3 Topic: Chemical Reactions II CR2-LO1

$\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$, $\Delta H = -2220 \text{ kJ}$. Heat released by 0.250 mol of propane is _____.

- A. 555 kJ
- B. 2220 kJ
- C. 8880 kJ
- D. 1110 kJ

A4 Topic: Chemical Reactions II CR2-LO2

On an energy diagram, reactants are at 40 kJ, the peak at 110 kJ and products at 90 kJ. E_a (forward) and ΔH are:

- A. 70 kJ and +50 kJ
- B. 110 kJ and +50 kJ
- C. 70 kJ and -50 kJ
- D. 20 kJ and +90 kJ

A5 Topic: Chemical Reactions II CR2-LO3

Which is **not** a requirement of collision theory for a reaction to occur?

- A. The particles must collide
- B. The collision must have energy $\geq E_a$
- C. The particles must collide in a suitable orientation
- D. The reaction must be exothermic

A6 Topic: Chemical Reactions II CR2-LO4

Which change will **not** increase the rate of reaction between zinc granules and dilute HCl?

- A. Using zinc powder
- B. Warming the acid
- C. Using a larger volume of the same acid
- D. Using more concentrated acid

A7 Topic: Chemical Reactions II CR2-LO4

A tablet label says 'Store below 25 °C in a dry place'. The main reason is to _____.

- A. increase E_a
- B. slow the degradation reactions of the drug
- C. make the reaction more exothermic
- D. shift the drug's equilibrium to the right

A8 Topic: Chemical Reactions II CR2-LO5

A catalyst increases the rate of a reaction because it _____.

- A. increases the kinetic energy of the particles
- B. provides an alternative pathway with a lower activation energy
- C. increases ΔH
- D. increases the concentration of reactants

A9 Topic: Chemical Reactions II CR2-LO5

Which statement about enzymes is correct?

- A. They are used up in the reactions they catalyse
- B. They work fastest at about 80 °C
- C. Each enzyme acts only on specific substrates that fit its active site
- D. They change the ΔH of the reaction

A10 Topic: Chemical Reactions II CR2-LO6

For $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$, $\Delta H = -198 \text{ kJ}$, which change shifts the equilibrium to the **right**?

- A. Raising the temperature
- B. Removing O_2
- C. Increasing the pressure
- D. Adding a catalyst

Part B · Structured questions

Show all working. The marks and Bloom's level are shown for each question.

Question B1 Topic: Chemical Reactions II CR2-LO1 5 marks, C3

Propane is the fuel in gas cylinders: $\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\ell)$, $\Delta H = -2220 \text{ kJ}$. (a) State whether the reaction is exothermic or endothermic, with a reason. (1 mark) (b) Calculate the heat released when 11.0 g of propane burns ($\text{C} = 12.01$, $\text{H} = 1.01$). (3 marks) (c) Write ΔH for $6\text{CO}_2(\text{g}) + 8\text{H}_2\text{O}(\ell) \rightarrow 2\text{C}_3\text{H}_8(\text{g}) + 10\text{O}_2(\text{g})$. (1 mark)

Question B2 Topic: Chemical Reactions II CR2-LO2 6 marks, C3

An energy diagram for a reaction shows the reactants at 80 kJ, the transition state at 200 kJ and the products at 140 kJ. (a) Calculate the activation energy of the forward reaction. (1 mark) (b) Calculate ΔH and classify the reaction. (2 marks) (c) Calculate the activation energy of the reverse reaction. (1 mark) (d) Sketch the energy diagram, labelling E_a and ΔH , and add a dashed line for the same reaction with a catalyst. (2 marks)

Question B3 Topic: Chemical Reactions II CR2-LO3 5 marks, C2

(a) State the three conditions that collision theory requires for a reaction to take place. (3 marks) (b) A mixture of hydrogen and oxygen gases can be stored for years, but it explodes when a spark is applied. Explain this using collision theory. (2 marks)

Question B4 Topic: Chemical Reactions II CR2-LO4 6 marks, C3

Hydrogen peroxide decomposes: $2\text{H}_2\text{O}_2(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\ell) + \text{O}_2(\text{g})$. (a) The concentration of H_2O_2 falls from 0.80 M to 0.50 M in 60 s. Calculate the average rate of loss of H_2O_2 and the rate of formation of O_2 . (3 marks) (b) Explain, using collision theory, why a pharmacist stores insulin in a refrigerator at 2–8 °C. (2 marks) (c) Explain why H_2O_2 fizzes immediately when applied to a cut. (1 mark)

Question B5 Topic: Chemical Reactions II CR2-LO6 5 marks, C3

Colourless N_2O_4 and brown NO_2 are in equilibrium in a sealed tube: $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$, $\Delta H = +58 \text{ kJ}$. (a) Define dynamic equilibrium. (1 mark) (b) Using Le Châtelier's principle, predict and explain the direction of shift when: (i) more NO_2 is added; (ii) the tube is placed in hot water; (iii) the pressure is increased; (iv) a catalyst is added. (4 marks)

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Adapted from *The Basics of GOB Chemistry* (Ball, Hill & Scott), LibreTexts, CC BY-NC-SA 3.0, sections 6.S and 7.4, with rates and equilibrium expanded from *Introductory Chemistry* (LibreTexts), chapter 15. Shared under the same licence for non-commercial teaching.

Periodic table · each box: atomic number, symbol, atomic mass, **electronegativity** · colours: metal, metalloid, nonmetal, noble gas

1A 1	2A 2	3	4	5	6	7	8	9	10	11	12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01 2.1																	2 He 4.00 —
3 Li 6.94 1.0	4 Be 9.01 1.5											5 B 10.81 2.0	6 C 12.01 2.5	7 N 14.01 3.0	8 O 16.00 3.5	9 F 19.00 4.0	10 Ne 20.18 —
11 Na 22.99 0.9	12 Mg 24.31 1.2											13 Al 26.98 1.5	14 Si 28.09 1.8	15 P 30.97 2.1	16 S 32.07 2.5	17 Cl 35.45 3.0	18 Ar 39.95 —
19 K 39.10 0.8	20 Ca 40.08 1.0	21 Sc 44.96 1.3	22 Ti 47.87 1.5	23 V 50.94 1.6	24 Cr 52.00 1.6	25 Mn 54.94 1.5	26 Fe 55.84 1.8	27 Co 58.93 1.9	28 Ni 58.69 1.9	29 Cu 63.55 1.9	30 Zn 65.38 1.6	31 Ga 69.72 1.6	32 Ge 72.63 1.8	33 As 74.92 2.0	34 Se 78.97 2.4	35 Br 79.90 2.8	36 Kr 83.80 —
37 Rb 85.47 0.8	38 Sr 87.62 1.0	39 Y 88.91 1.2	40 Zr 91.22 1.4	41 Nb 92.91 1.6	42 Mo 95.95 1.8	43 Tc [98] 1.9	44 Ru 101.07 2.2	45 Rh 102.91 2.2	46 Pd 106.42 2.2	47 Ag 107.87 1.9	48 Cd 112.41 1.7	49 In 114.82 1.7	50 Sn 118.71 1.8	51 Sb 121.76 1.9	52 Te 127.60 2.1	53 I 126.90 2.5	54 Xe 131.29 —
55 Cs 132.91 0.7	56 Ba 137.33 0.9	57–71	72 Hf 178.49 1.3	73 Ta 180.95 1.5	74 W 183.84 1.7	75 Re 186.21 1.9	76 Os 190.23 2.2	77 Ir 192.22 2.2	78 Pt 195.08 2.2	79 Au 196.97 2.4	80 Hg 200.59 1.9	81 Tl 204.38 1.8	82 Pb 207.2 1.9	83 Bi 208.98 1.9	84 Po [209] 2.0	85 At [210] 2.2	86 Rn [222] —
87 Fr [223] 0.7	88 Ra [226] 0.9	89–103	104 Rf [267] —	105 Db [268] —	106 Sg [269] —	107 Bh [270] —	108 Hs [277] —	109 Mt [278] —	110 Ds [281] —	111 Rg [282] —	112 Cn [285] —	113 Nh [286] —	114 Fl [289] —	115 Mc [290] —	116 Lv [293] —	117 Ts [294] —	118 Og [294] —
Lanthanides		57 La 138.91 —	58 Ce 140.12 —	59 Pr 140.91 —	60 Nd 144.24 —	61 Pm [145] —	62 Sm 150.36 —	63 Eu 151.96 —	64 Gd 157.25 —	65 Tb 158.93 —	66 Dy 162.50 —	67 Ho 164.93 —	68 Er 167.26 —	69 Tm 168.93 —	70 Yb 173.05 —	71 Lu 174.97 —	
Actinides		89 Ac [227] 1.1	90 Th 232.04 1.3	91 Pa 231.04 1.4	92 U 238.03 1.4	93 Np [237] —	94 Pu [244] —	95 Am [243] —	96 Cm [247] —	97 Bk [247] —	98 Cf [251] —	99 Es [252] —	100 Fm [257] —	101 Md [258] —	102 No [259] —	103 Lr [262] —	

Atomic masses to 2 d.p. ([n] = mass number of the most stable isotope). Electronegativity: Pauling values as in Ball, *The Basics of GOB Chemistry*, Figure 4.4.3 (La–Lu 1.0–1.2; noble gases have no value).

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Part A: 10 multiple-choice questions (1 mark each). Part B: 5 structured questions (27 marks). Same format as the final exam.

Answer key

Part A · Multiple choice

A1 Topic: Chemical Reactions II CR2-LO1

Which statement about an endothermic reaction is correct?

- A. ΔH is negative and the surroundings get warmer
- B. ΔH is positive and heat is absorbed from the surroundings**
- C. Bond forming releases more energy than bond breaking absorbs
- D. It can never happen at room temperature

Endothermic: heat absorbed, $\Delta H > 0$, surroundings cool (e.g. a cold pack).

A2 Topic: Chemical Reactions II CR2-LO1

$\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$, $\Delta H = +180.6 \text{ kJ}$. What is ΔH for $4\text{NO} \rightarrow 2\text{N}_2 + 2\text{O}_2$?

- A. $+361.2 \text{ kJ}$
- B. -180.6 kJ
- C. -361.2 kJ**
- D. $+90.3 \text{ kJ}$

Reverse: change the sign (-180.6 kJ); double: -361.2 kJ .

A3 Topic: Chemical Reactions II CR2-LO1

$\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$, $\Delta H = -2220 \text{ kJ}$. Heat released by 0.250 mol of propane is _____.

A. 555 kJ

B. 2220 kJ

C. 8880 kJ

D. 1110 kJ

$0.250 \text{ mol} \times 2220 \text{ kJ/mol} = 555 \text{ kJ}$.

A4 Topic: Chemical Reactions II CR2-LO2

On an energy diagram, reactants are at 40 kJ, the peak at 110 kJ and products at 90 kJ. E_a (forward) and ΔH are:

A. 70 kJ and +50 kJ

B. 110 kJ and +50 kJ

C. 70 kJ and -50 kJ

D. 20 kJ and +90 kJ

$E_a = 110 - 40 = 70 \text{ kJ}$; $\Delta H = 90 - 40 = +50 \text{ kJ}$.

A5 Topic: Chemical Reactions II CR2-LO3

Which is **not** a requirement of collision theory for a reaction to occur?

A. The particles must collide

B. The collision must have energy $\geq E_a$

C. The particles must collide in a suitable orientation

D. The reaction must be exothermic

Endothermic reactions also occur by effective collisions.

A6 Topic: Chemical Reactions II CR2-LO4

Which change will **not** increase the rate of reaction between zinc granules and dilute HCl?

- A. Using zinc powder
- B. Warming the acid
- C. Using a larger volume of the same acid**
- D. Using more concentrated acid

Same concentration means the same collision frequency per unit volume; extra volume does not speed it up.

A7 Topic: Chemical Reactions II CR2-LO4

A tablet label says 'Store below 25 °C in a dry place'. The main reason is to _____.

- A. increase E_a
- B. slow the degradation reactions of the drug**
- C. make the reaction more exothermic
- D. shift the drug's equilibrium to the right

Lower temperature and less moisture reduce the rate of breakdown.

A8 Topic: Chemical Reactions II CR2-LO5

A catalyst increases the rate of a reaction because it _____.

- A. increases the kinetic energy of the particles
- B. provides an alternative pathway with a lower activation energy**
- C. increases ΔH
- D. increases the concentration of reactants

It lowers E_a ; ΔH and the products are unchanged.

A9 Topic: Chemical Reactions II CR2-LO5

Which statement about enzymes is correct?

- A. They are used up in the reactions they catalyse
- B. They work fastest at about 80 °C
- C. Each enzyme acts only on specific substrates that fit its active site**
- D. They change the ΔH of the reaction

Enzymes are specific, reusable and denature at high temperature.

A10 Topic: Chemical Reactions II CR2-LO6

For $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$, $\Delta H = -198 \text{ kJ}$, which change shifts the equilibrium to the **right**?

- A. Raising the temperature
- B. Removing O_2
- C. Increasing the pressure**
- D. Adding a catalyst

3 mol of gas on the left, 2 on the right: higher pressure favours fewer gas molecules.

Part B · Structured questions

Question B1 Topic: Chemical Reactions II CR2-LO1 5 marks, C3

Propane is the fuel in gas cylinders: $\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\ell)$, $\Delta H = -2220 \text{ kJ}$. (a) State whether the reaction is exothermic or endothermic, with a reason. (1 mark) (b) Calculate the heat released when 11.0 g of propane burns ($\text{C} = 12.01$, $\text{H} = 1.01$). (3 marks) (c) Write ΔH for $6\text{CO}_2(\text{g}) + 8\text{H}_2\text{O}(\ell) \rightarrow 2\text{C}_3\text{H}_8(\text{g}) + 10\text{O}_2(\text{g})$. (1 mark)

(a) **Exothermic**, because ΔH is negative (heat is released). (b) $M(\text{C}_3\text{H}_8) = 3(12.01) + 8(1.01) = 44.11 \text{ g/mol}$. Moles = $11.0 \div 44.11 = 0.249 \text{ mol}$. Heat = $0.249 \times 2220 = \mathbf{554 \text{ kJ released}}$. (c) Reversed and doubled: **$\Delta H = +4440 \text{ kJ}$** .

Step	Marking criteria	Marks
1	Exothermic, because ΔH is negative	1
2	Molar mass 44.11 g/mol	1
3	Moles = 0.249 mol	1
4	Heat = 554 kJ released (accept 553–555)	1
5	$\Delta H = +4440 \text{ kJ}$	1
	Total	5

Question B2 Topic: Chemical Reactions II CR2-LO2 6 marks, C3

An energy diagram for a reaction shows the reactants at 80 kJ, the transition state at 200 kJ and the products at 140 kJ. (a) Calculate the activation energy of the forward reaction. (1 mark) (b) Calculate ΔH and classify the reaction. (2 marks) (c) Calculate the activation energy of the reverse reaction. (1 mark) (d) Sketch the energy diagram, labelling E_a and ΔH , and add a dashed line for the same reaction with a catalyst. (2 marks)

(a) $E_a = 200 - 80 = \mathbf{120\text{ kJ}}$. (b) $\Delta H = 140 - 80 = \mathbf{+60\text{ kJ}}$, so the reaction is **endothermic**. (c) $E_a(\text{reverse}) = 200 - 140 = \mathbf{60\text{ kJ}}$. (d) Products drawn higher than reactants; one hill; E_a arrow from reactants to peak and ΔH arrow from reactants to products; the catalysed path is a lower hill with the same start and end levels.

Step	Marking criteria	Marks
1	$E_a = 120\text{ kJ}$	1
2	$\Delta H = +60\text{ kJ}$	1
3	Endothermic	1
4	$E_a(\text{reverse}) = 60\text{ kJ}$	1
5	Diagram with products above reactants and E_a , ΔH labelled	1
6	Catalysed path: lower peak, same reactant and product levels	1
	Total	6

Question B3 Topic: Chemical Reactions II CR2-LO3 5 marks, C2

(a) State the three conditions that collision theory requires for a reaction to take place. (3 marks) (b) A mixture of hydrogen and oxygen gases can be stored for years, but it explodes when a spark is applied. Explain this using collision theory. (2 marks)

(a) The particles must **collide**; the collision must have energy $\geq E_a$; the particles must meet in the **correct orientation**. (b) At room temperature almost no collisions have energy $\geq E_a$, so the rate is negligible. The spark gives some molecules enough energy to react; the exothermic reaction then heats nearby molecules so more and more collisions become effective, giving an explosion.

Step	Marking criteria	Marks
1	Particles must collide	1
2	With energy $\geq$ activation energy	1
3	In the correct orientation	1
4	Room temperature: too few collisions have energy $\geq E_a$	1
5	Spark supplies E_a ; heat released makes more collisions effective	1
	Total	5

Question B4 Topic: Chemical Reactions II CR2-LO4 6 marks, C3

Hydrogen peroxide decomposes: $2\text{H}_2\text{O}_2(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\ell) + \text{O}_2(\text{g})$. (a) The concentration of H_2O_2 falls from 0.80 M to 0.50 M in 60 s. Calculate the average rate of loss of H_2O_2 and the rate of formation of O_2 . (3 marks) (b) Explain, using collision theory, why a pharmacist stores insulin in a refrigerator at 2–8 °C. (2 marks) (c) Explain why H_2O_2 fizzes immediately when applied to a cut. (1 mark)

(a) Change = $0.80 - 0.50 = 0.30$ M. Rate = $0.30 \div 60 = 5.0 \times 10^{-3}$ M/s. O_2 forms at half this rate: 2.5×10^{-3} M/s. (b) At low temperature the molecules move more slowly, so they collide less often and a smaller fraction of collisions have energy $\geq E_a$. The degradation reactions of insulin are slower, so it stays effective for longer. (c) The enzyme **catalase** in blood is a catalyst that lowers E_a , so decomposition is very fast.

Step	Marking criteria	Marks
1	Change in concentration 0.30 M	1
2	Rate = 5.0×10^{-3} M/s	1
3	O_2 rate = 2.5×10^{-3} M/s	1
4	Fewer collisions / slower particles	1
5	Smaller fraction with energy $\geq E_a$, so slower degradation	1
6	Catalase (enzyme) lowers E_a	1
	Total	6

Question B5 Topic: Chemical Reactions II CR2-LO6 5 marks, C3

Colourless N_2O_4 and brown NO_2 are in equilibrium in a sealed tube: $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$, $\Delta H = +58 \text{ kJ}$. (a) Define dynamic equilibrium. (1 mark) (b) Using Le Châtelier's principle, predict and explain the direction of shift when: (i) more NO_2 is added; (ii) the tube is placed in hot water; (iii) the pressure is increased; (iv) a catalyst is added. (4 marks)

(a) The forward and reverse reactions continue at **equal rates**, so concentrations stay constant. (b)(i) **Left**: the system uses up added NO_2 . (ii) **Right**: the forward reaction is endothermic, so added heat is absorbed by forming more NO_2 (the gas turns browner). (iii) **Left**: 1 mol of gas on the left, 2 on the right; higher pressure favours fewer gas molecules. (iv) **No shift**: a catalyst speeds up both directions equally.

Step	Marking criteria	Marks
1	Forward rate = reverse rate; concentrations constant	1
2	(i) left	1
3	(ii) right, endothermic forward reaction	1
4	(iii) left, fewer moles of gas	1
5	(iv) no shift	1
	Total	5

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